

CMPE 58I

Human-inspired Machine Intelligence

[2025-2026 Fall]

*Impact of an **anti**-biomimetic training regimen based on early visual experience on neural network organization and behavior*

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03.01.2026



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Our Work

anti-biomimetic



Impact of a biomimetic training regimen based on early visual experience on neural network organization and behavior

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Terminology

Biomimetic training for deep learning refers to training methodologies that replicate the development, plasticity, or structural learning processes of biological organisms.

Mammalian visual pathway cells can be broadly categorized into [*]:

- **Magnocellular**
 - Low spatial frequency (motion, gross depth)
 - Low chromatic sensitivity (black-white perception)
- **Parvocellular**
 - High spatial frequency (texture shape and finer depth)
 - High chromatic sensitivity (color perception)



Overall Problem

Deep neural networks are powerful tools for wide variety of tasks

However,

- **Biased towards texture** while humans rely on on global shape for classification decisions [*]
- **Lack robustness** against perturbations that degrade input quality [**]

*Geirhos et al., 2018a

**Geirhos et al.,
2018b, 2020, Taori et
al., 2020, Dunn et al.,
2021.

Research Question

1. How does (anti)biomimetic training impact **shape & texture bias** of deep learning systems?
2. Could the joint development/decline of **spatial frequency** and **color sensitivity** be the driving force behind the organization of the parvocellular and magnocellular pathways?
 - a. *This question is motivated by the fact that human infants initially experience the world through blurry, achromatic vision, which only gradually matures into high-acuity, full-color perception.[*,**]*

* Russell J Adams and Mary L Courage. A psychophysical test of the early maturation of infants' mid-and long-wavelength retinal cones. *Infant Behavior and Development*, 25(2):247–254, 2002.

**Velma Dobson and Davida Y Teller. Visual acuity in human infants: a review and comparison of behavioral and electrophysiological studies. *Vision research*, 18(11):1469–1483, 1978.

Proposed Approach

Baseline Work

Standard regimen (Control):

- High-resolution, full-color images for entire training duration

Biomimetic regimen:

- First half: Blurry, achromatic (grayscale) images
- Second half: High-resolution, full-color images

Proposed Approach

Our Work

Standard regimen:

- High-resolution, full-color images for entire training duration

Biomimetic regimen:

- First half: Blurry, achromatic (grayscale) images
- Second half: High-resolution, full-color images

Anti-Biomimetic regimen:

- First half: High-resolution, full-color images
- Second half: Blurry, achromatic (grayscale) images

Evaluation: Phase I

Shape vs. Texture Bias

Evaluation: Phase I

Methodology: Shape vs. Texture Bias

Assume we trained the model with the suggested regimen.

How do we find whether it is biased towards shape or texture?

Evaluation: Phase I

Methodology: Shape vs. Texture Bias

Assume we trained the model with the suggested regimen.

How do we find whether it is biased towards shape or texture?

WE CAN USE A
SPECIAL DATASET

Evaluation: Phase I

Dataset: Shape vs. Texture Bias

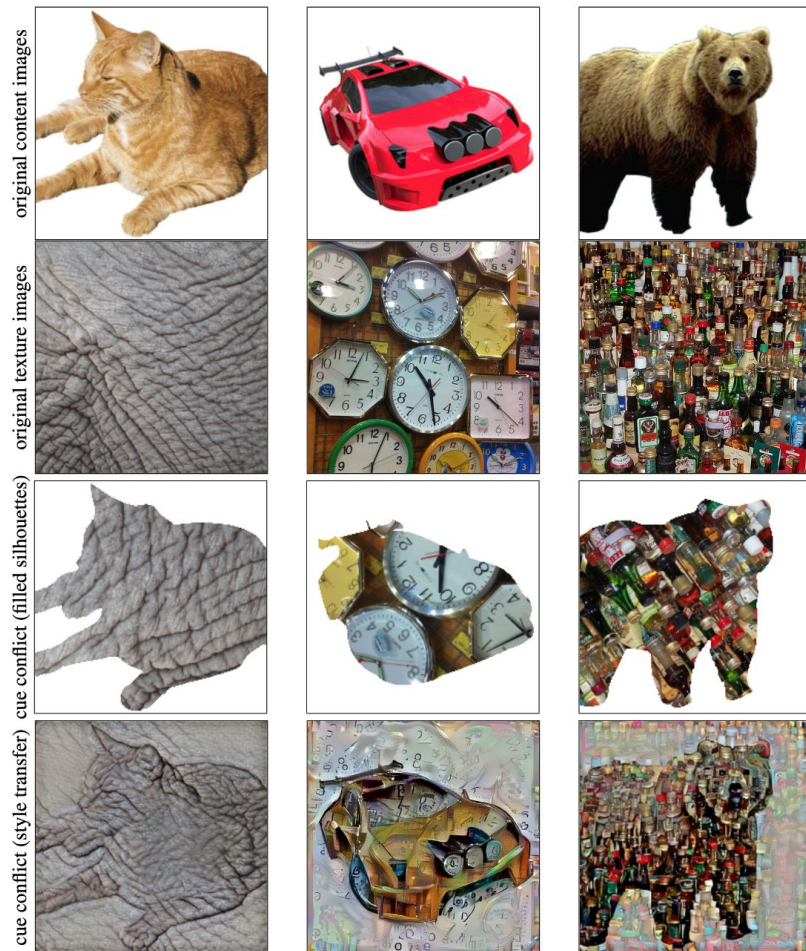
Cue-conflict subset

1280 images

Categorize decisions as:

- [Category 1, Category 2, Other]

IMAGENET-TRAINED CNNs ARE BIASED TOWARDS
TEXTURE; INCREASING SHAPE BIAS IMPROVES
ACCURACY AND ROBUSTNESS



Disclaimer

Only **AlexNet** was used in evaluations since original study focused only on this architecture for receptive field analyses. See appendix for other architectures such as Inception and ResNet.

Technical Limitations

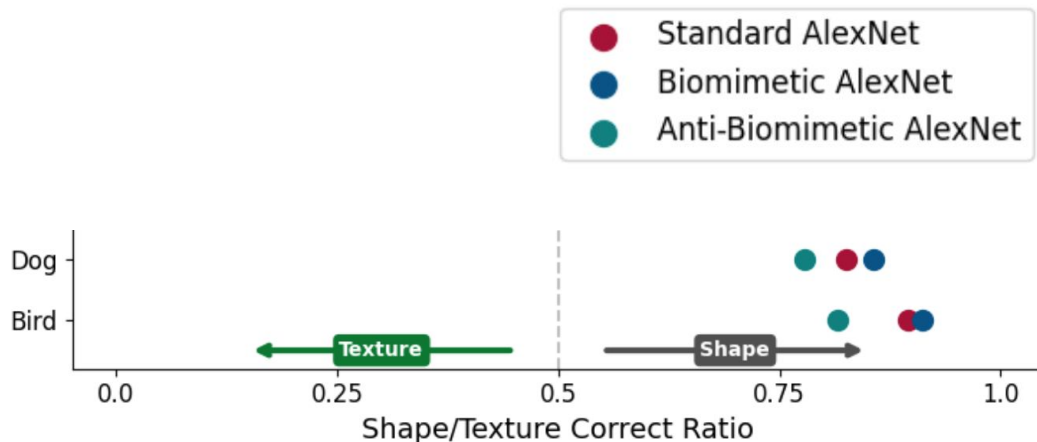
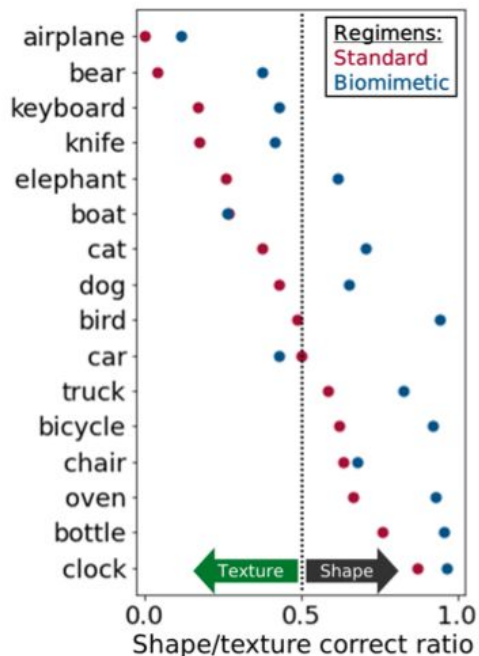
- Baseline work's **unofficial code** was written with **Tensorflow** which was re-written with **PyTorch**.
- Data Usage: 40% Training / 30% Validation sets utilized. (originally planned 100 / 100).
- Training: 120 epochs (originally planned: 200).

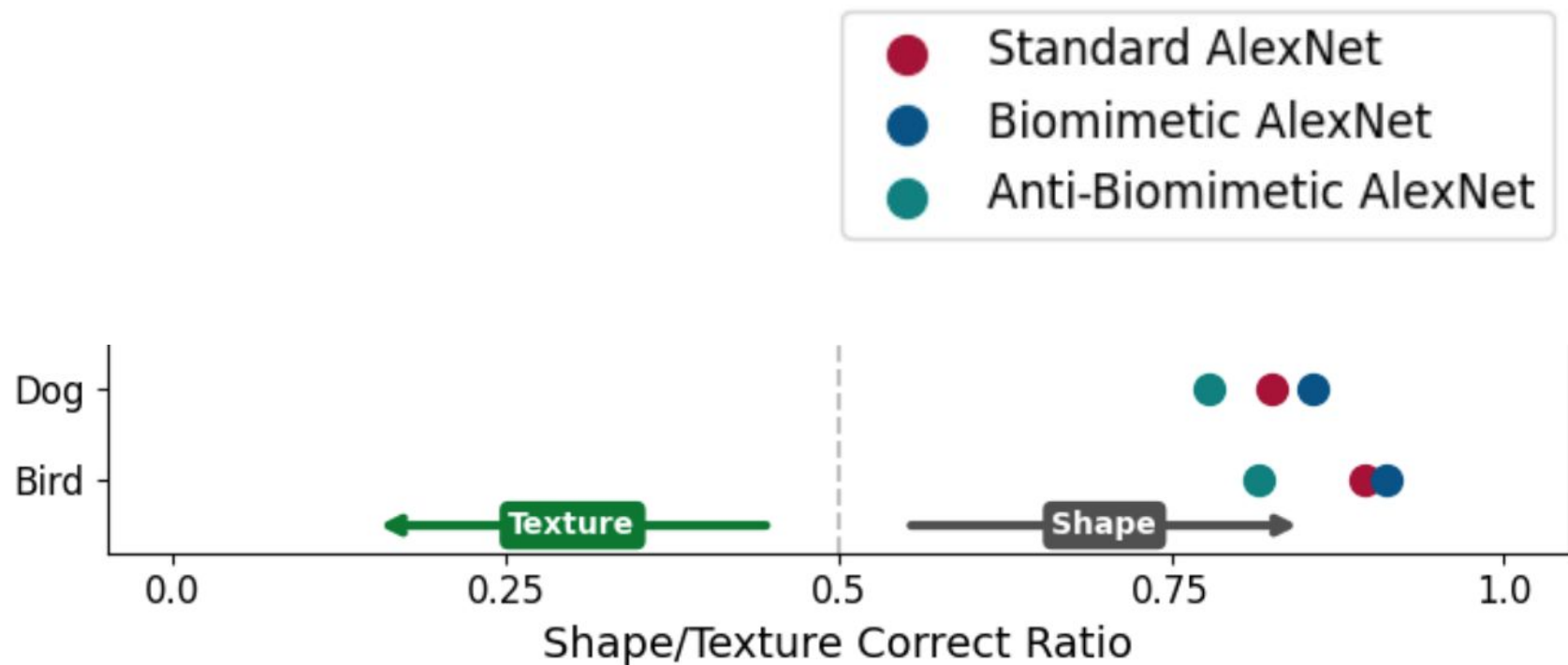
Impact: Meaningful results were achieved; however, the reduced **scale** may have dampened the visibility of more subtle trends or drastic observations.

Evaluation Phase I

Shape vs. Texture Bias Scatter Plot

In order of original study vs. ours

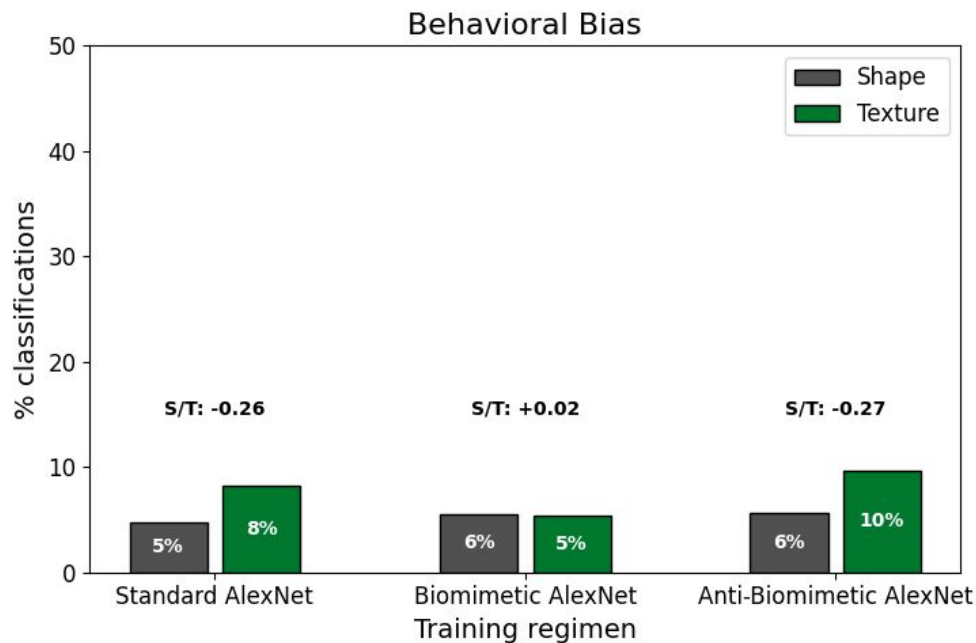
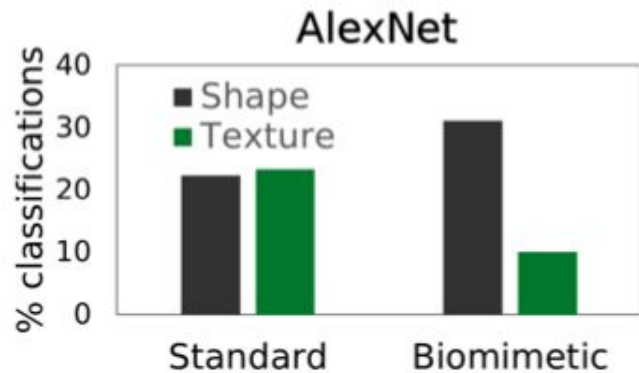




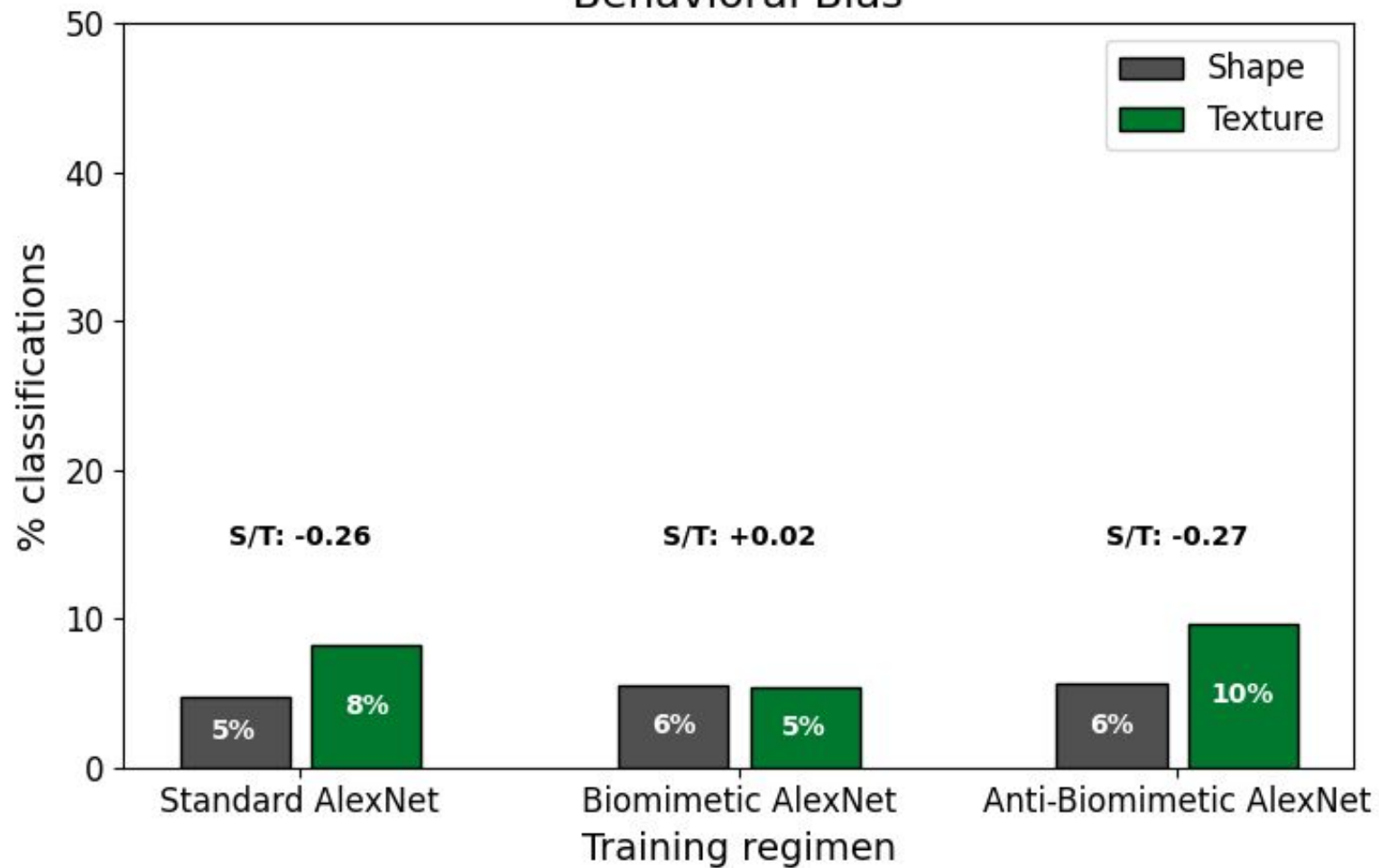
Evaluation: Phase I

Shape vs. Texture Bias Bar Plot

In order of original study vs. ours



Behavioral Bias



Evaluation: Phase II

Receptive Field Analyses

Evaluation: Phase II

Methodology: Receptive Field Analyses

We want to see whether **joint development/decline of spatial frequency and color sensitivity** drives the force behind the **organization of the parvocellular and magnocellular pathways**

Evaluation: Phase II

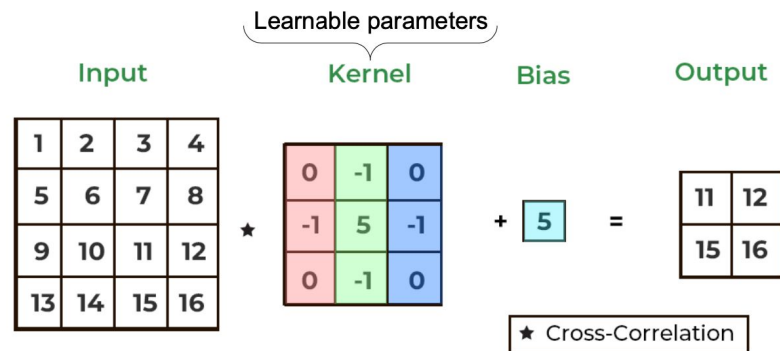
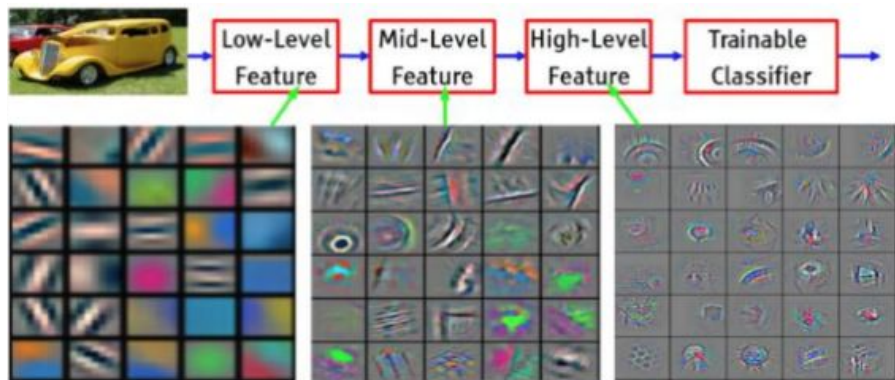
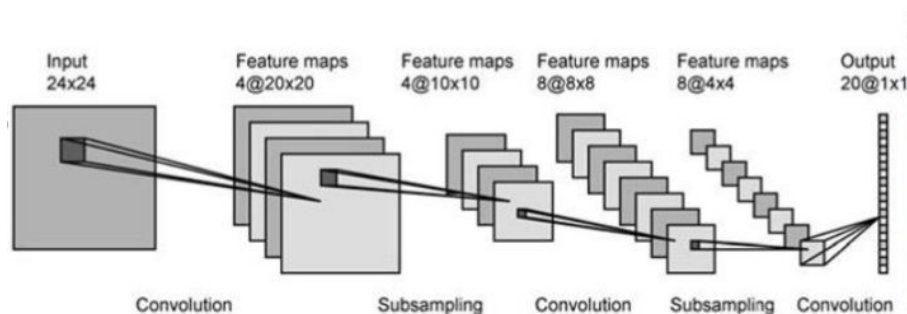
Methodology: Receptive Field Analyses

We want to see whether **joint development/decline of spatial frequency and color sensitivity** drives the force behind the **organization of the parvocellular and magnocellular pathways**

We have **joint development/decline of spatial frequency and color sensitivity** covered with our training regimen

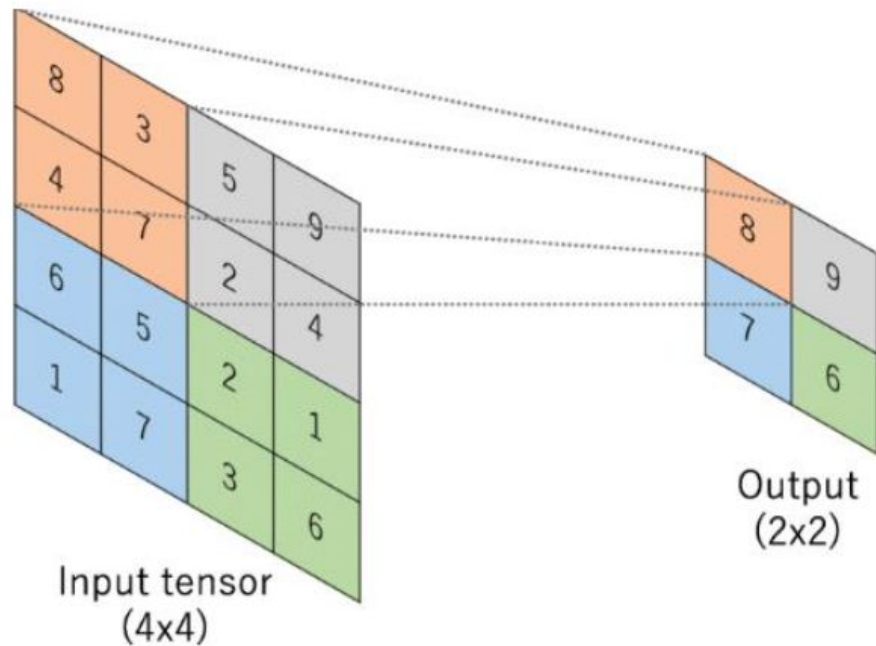
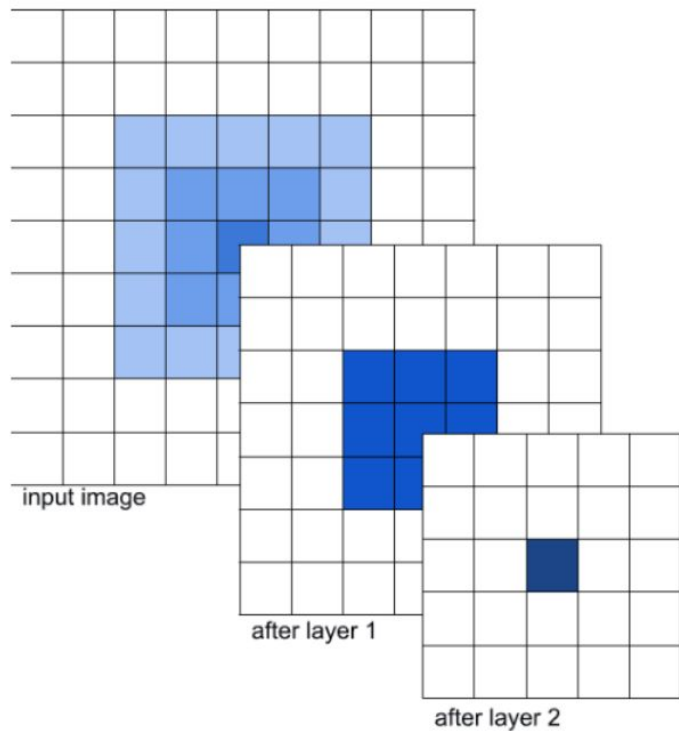
Terminology

Convolutional Neural Networks



Terminology

Receptive Field



Modified AlexNet

Design Choices

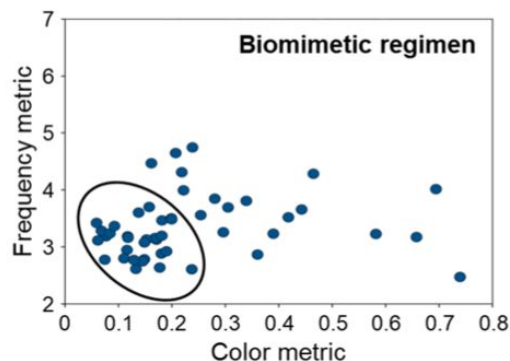
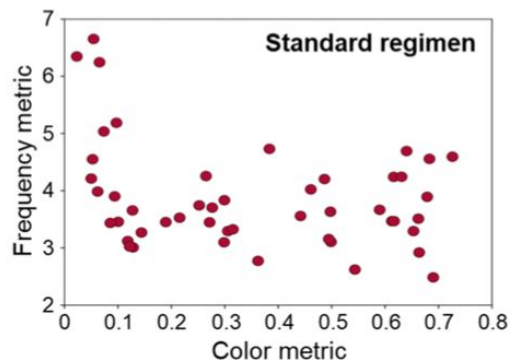
To ensure the filters were large enough for precise frequency analysis, they adjusted the architecture to feature **fewer but larger filters**: the first layer contained **48 RFs**, each with a size of **22x22** pixels.

- **Similar results** were obtained with **original architecture**. However, the modified architecture resulted in easier-to-interpret color vs. frequency scatter plots.

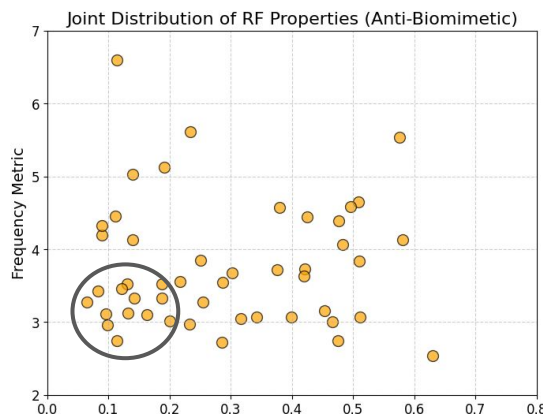
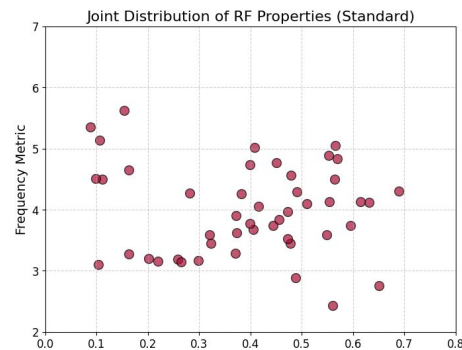
Evaluation: Phase II

Spatial Frequency of Individual RFs

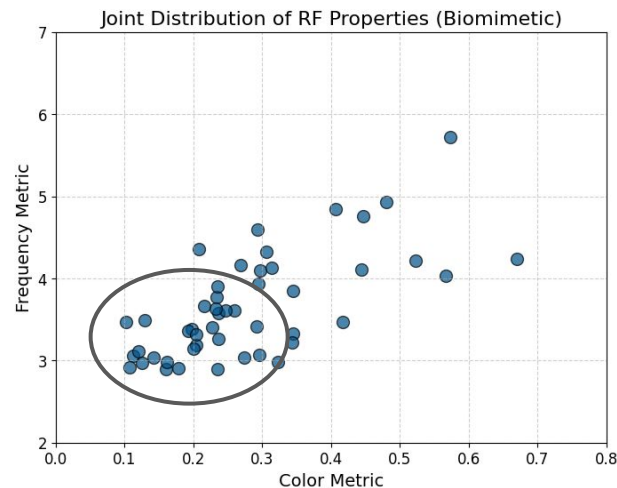
In order of original study vs. ours



Standard



Biomimetic

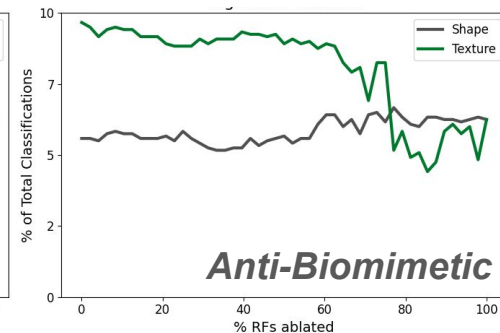
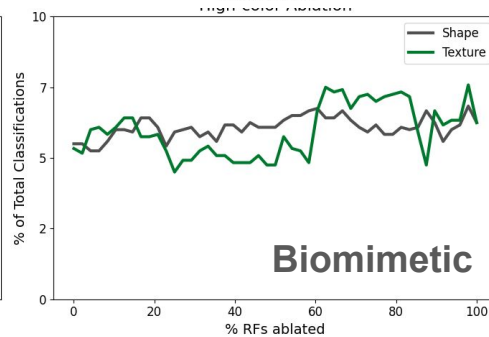
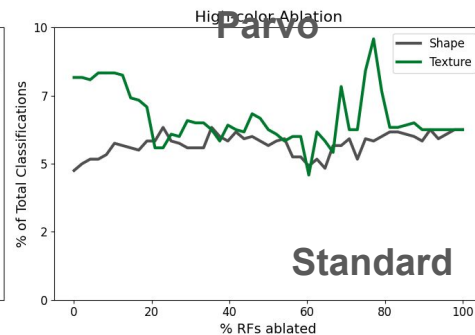
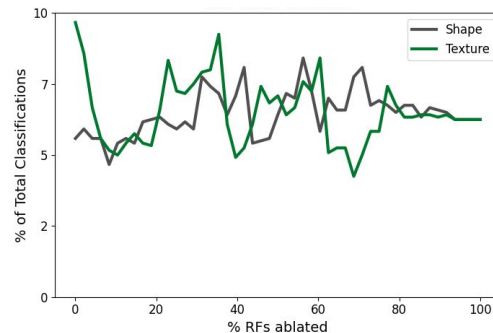
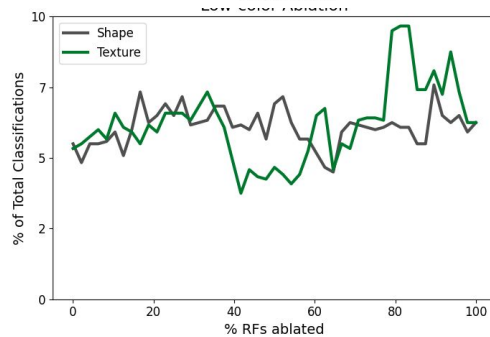
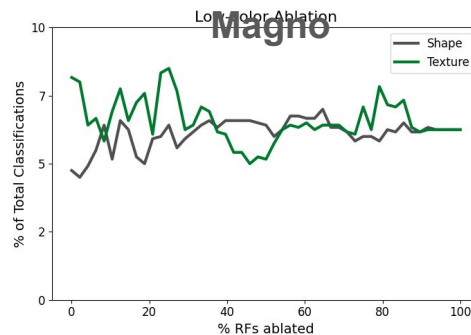
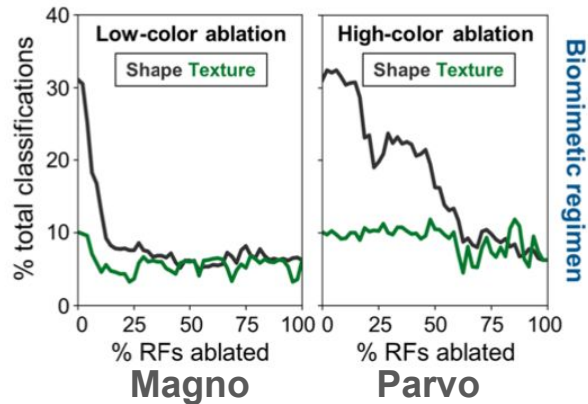
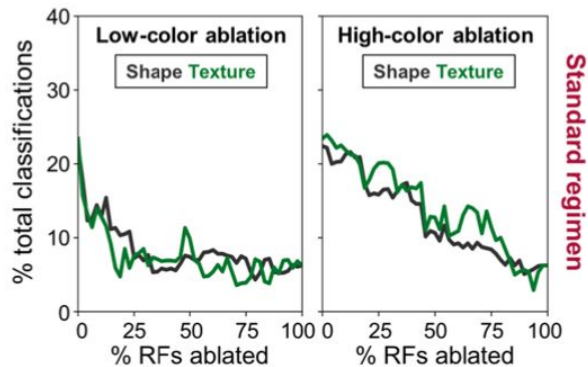


Anti-Biomimetic

Evaluation: Phase II

Colorfulness of Individual RFs

In order of original study vs. ours



Conclusion

1. Verification of the Parvo/Magno Mechanism

Our results confirm that the organization of the visual pathway is driven by the **temporal order** of sensory experience, not just the data itself.

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Our results confirm that the organization of the visual pathway is driven by the **temporal order** of sensory experience, not just the data itself.

2. Why did the Anti-Biomimetic model end up with higher texture bias than the Standard model?

Introducing blur in the second half was **too late to trigger a reorganization**. Instead of building a global shape bias (Magno), the network likely "doubled down" on its **existing texture detectors** to survive the degradation.

Conclusion

1. Verifying

Our results show that the temporal dynamics of learning are critical for the development of global shape selectivity.

2. Why the Stages?

Introduction of building blocks of existing visual systems.

There is a **critical period** for learning global shape.

If the network is not forced to rely on coarse, achromatic information (**Magno features**) at the very beginning of training,

it will default to a reliance on local texture (**Parvo features**), regardless of subsequent training conditions.



Thank you for your attention

For additional information, please refer to the references and appendix sections at the end of this presentation.

References

Margaret Livingstone and David Hubel. Segregation of form, color, movement, and depth: anatomy, physiology, and perception. *Science*, 240(4853):740–749, 1988.

Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A Wichmann, and Wieland Brendel. Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness. arXiv preprint arXiv:1811.12231, 2018a.

Robert Geirhos, Carlos RM Temme, Jonas Rauber, Heiko H Schütt, Matthias Bethge, and Felix A Wichmann. Generalisation in humans and deep neural networks. *Advances in neural information processing systems*, 31, 2018b.

Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.

Rohan Taori, Achal Dave, Vaishaal Shankar, Nicholas Carlini, Benjamin Recht, and Ludwig Schmidt. Measuring robustness to natural distribution shifts in image classification. *Advances in Neural Information Processing Systems*, 33:18583–18599, 2020.

Isaac Dunn, Hadrien Pouget, Daniel Kroening, and Tom Melham. Exposing previously undetectable faults in deep neural networks. In *Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis*, pages 56–66, 2021.

Appendix

Training Setup

Baseline Work (Biomimetic Regimen)

Dataset: ImageNet

Hyperparameters:

- 200 epochs
 - First 100 epochs: blurry, achromatic
 - Last 100 epochs: full-color, high-res
- Constant learning rate: 0.001
- Batch size: 128
- Loss: Categorical cross-entropy
- Optimizer: Stochastic Gradient Descent with Nesterov Momentum of 0.9

Training Setup

Our Work (Anti-Biomimetic Regimen)

Dataset: ImageNet

Hyperparameters:

- Data Usage: 40% Training / 30% Validation sets utilized.
- 120 epochs
 - First 60 epochs: full-color, high-resolution
 - Last 60 epochs: blurry, achromatic
- Constant learning rate: 0.001
- Batch size: 128
- Loss: Categorical cross-entropy
- Optimizer: Stochastic Gradient Descent with Nesterov Momentum of 0.9

Architecture & Data Preprocessing

Data preprocessing:

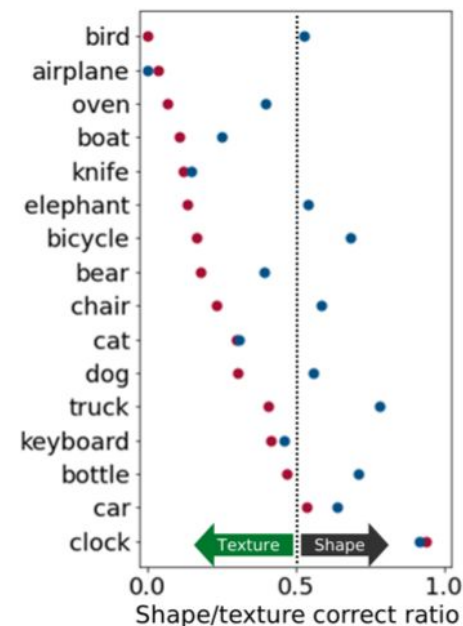
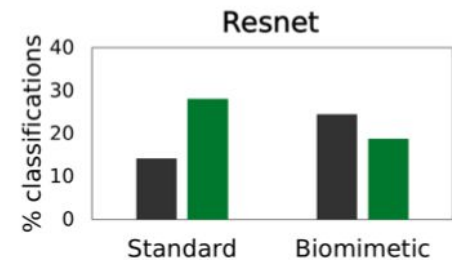
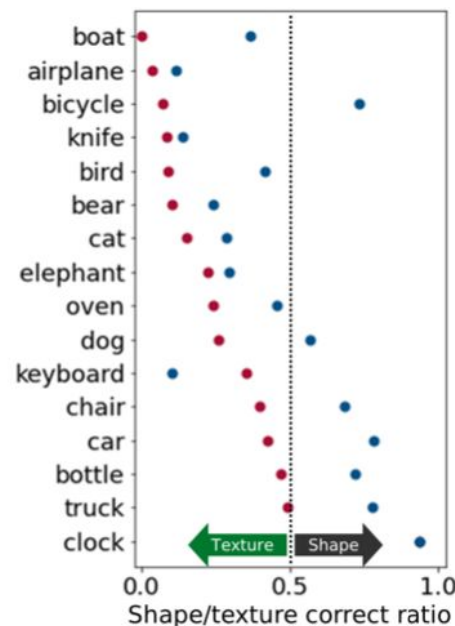
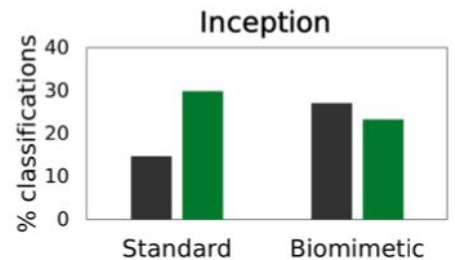
- Training
 - Random crop from 256x256 to 227x227
 - Horizontal flipping at random
 - Pixel value rescaling from $[0, 255]$ to $[-1, 1]$

Modified architecture:

- AlexNet with larger receptive fields (RF) in first convolutional layer
 - 48 instead of 96, each increased from 11x11 to 22x22

Shape vs. Texture Bias for different architectures

Baseline Work



Color metric for receptive field analysis

$$x = R \cos(0^\circ) + G \cos(120^\circ) + B \cos(-120^\circ)$$

$$y = R \sin(0^\circ) + G \sin(120^\circ) + B \sin(-120^\circ)$$

$$m = \sqrt{x^2 + y^2}$$

Spatial frequency metric for receptive field analysis

$$\text{weighted average frequency} = \frac{\sum_f (\text{amp}(f) \cdot f)}{\sum_f \text{amp}(f)}$$

where *amp* refers to the amplitude of a given frequency, and *f* refers to the frequency itself.